# Short Notes & Keywords

Engineering Chemistry (UE26CY151A) · Semester 1 · Unit 1 · Compiled from PESU AV-summary transcripts, class slides, the Unit-1 slide PDF and past ESA papers · 13 Oct 2026

★★★ high · ★★ medium · ★ low/conceptual weight | Not on current slides Transcript only Added for clarity | source tags: [S004·8] slide, [T005] transcript, [PDF] 137-page deck, [ESA] past paper

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# M1 · Molecular Spectroscopy

## T1 · EM radiation & interaction with matter

Spectroscopy studies radiation–matter interaction. A spectrophotometer has source, monochromator, sample, detector, recorder. EM waves have E ⊥ B ⊥ direction. $E = h\nu$, $\nu = c/\lambda$, $\bar{\nu} = 1/\lambda$. Atomic spectra are lines; molecular spectra are bands.

## T2 · Electromagnetic spectrum & regions

Radio → NMR/ESR; microwave → rotation; IR → vibration; UV-Vis → valence electrons; X-ray → inner electrons; γ → nuclei. Raman: Rayleigh (no shift), Stokes (lower ν), anti-Stokes (higher ν).

## T3 · Born–Oppenheimer approximation

$E = E_{rot} + E_{vib} + E_{elec}$, $E_{rot} \ll E_{vib} \ll E_{elec}$, components independent; translation negligible.

## T3b · Beer–Lambert law

$A = \log(I_0/I) = \varepsilon c l$ (older syllabus; not on current slides).

## T4 · Rotational (microwave) spectroscopy

Rigid rotor: two masses at fixed $r_0$. $I = \mu r_0^2$. $\varepsilon_J = BJ(J+1)$. Needs permanent dipole; $\Delta J = \pm 1$. Lines at $2B(J+1)$, spacing $2B$. From B get I and $r_0$.

## T5 · Vibrational (IR) – SHO

SHO: spring/Hooke's law, parabolic PE. $\varepsilon_v = (v+\tfrac{1}{2})\bar{\nu}$, ZPE = $\tfrac{1}{2}\bar{\nu}$, equal spacing, $\Delta v = \pm 1$ → one line at $\bar{\nu}$. Needs changing dipole; $\bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)}$.

## T6 · Vibrational – anharmonic

Morse potential: $\varepsilon_v = (v+\tfrac{1}{2})\bar{\omega_e} - (v+\tfrac{1}{2})^2\bar{\omega_e}x_e$. Levels converge; $\Delta v = \pm 1, \pm 2\ldots$; fundamental, overtones; hot bands left of fundamental.

## T7 · Electronic spectroscopy & Franck–Condon

HOMO → LUMO transitions; vibrational coarse structure gives progressions (v′,v″); no vibrational selection rule; Franck–Condon: vertical transitions → intensity pattern by $r_e'$ vs $r_e''$; continuum = dissociation.

# M2 · Phase Equilibria

## T8 · Basic terms & phase rule

Phase = homogeneous, separable part; component = min. independent constituents; F = min. intensive variables. F = C−P+2.

## T9 · Derivation of phase rule

Variables P(C−1)+2, constraints C(P−1), equal μ in all phases → F = C−P+2.

## T10 · One-component system: water

F = 3−P. Areas ice/water/vapour F=2; lines OA, OB, OC F=1; triple point O F=0 (0.0098 °C, 4.58 mmHg); critical 374 °C, 218 atm; OC slope negative; OA′ metastable.

## T11 · Two-component: reduced phase rule & Pb–Ag

Condensed system at fixed P: F = C−P+1. Pb–Ag eutectic: 303 °C, 2.6 % Ag, F=0; curves F=1.

## T12 · Thermal analysis / cooling curves

Cooling curves: halt for pure/eutectic, slope break for mixtures; plot to build phase diagram.

# M3 · Computational Chemistry

## T13 · Computational chemistry

Computational chemistry simulates molecules: MM/MD (classical), semi-empirical, ab initio, DFT (quantum). Cheap, safe; AI adds speed.

# Keyword index

| Keyword | Topic | Meaning |
|---|---|---|
| ab initio | T13 | wavefunction-based |
| absorption spectrum | T1 | radiation absorbed as molecule moves to higher level |
| anharmonicity constant $x_e$ | T6 | small positive constant in Morse levels |
| anti-Stokes line | T2 | Raman line at higher frequency |
| band spectrum | T1 | molecular spectrum from electronic+vib+rot transitions |
| bivariant | T8 | F = 2 |
| centre of mass | T4 | point where $m_1r_1 = m_2r_2$ |
| chemical potential | T9 | $(\partial G/\partial n_i)$; equal in all phases at equilibrium |
| Clausius–Clapeyron | T10 | $dP/dT = \Delta H/T\Delta V$ |
| coarse structure | T7 | vibrational structure on electronic bands |
| component | T8 | min. independent constituents |
| condensed system | T11 | no gas phase |
| continuum | T7 | non-quantised dissociation absorption |
| critical point | T10 | 374 °C, 218 atm |
| degrees of freedom | T8 | F |
| DFT | T13 | uses electron density |
| eutectic | T11 | lowest-melting mixture, F=0 |
| fine structure | T7 | rotational structure |
| force constant | T5 | k, N $m^{-1}$ |
| fundamental | T6 | v=0→1 |
| Gibbs phase rule | T8 | F = C−P+2 |
| gross selection rule | T4 | dipole requirement |
| HOMO / LUMO | T7 | frontier orbitals |
| Hooke's law | T5 | F = −kx |
| hot band | T6 | transition from v=1 |
| invariant | T8 | F = 0 |
| metastable | T10 | supercooled water, OA′ |
| molecular dynamics | T13 | atomic motion simulation |
| molecular mechanics | T13 | balls and springs |
| moment of inertia | T4 | $I = \mu r_o^2$ |
| Morse function | T6 | anharmonic PE |
| overtone | T6 | v=0→2, 3… |
| phase | T8 | homogeneous separable part |
| progression | T7 | series (v′,0) lines |
| Rayleigh scattering | T2 | elastic scattering |
| reduced mass | T4 | $\mu = m_1m_2/(m_1+m_2)$ |
| reduced phase rule | T11 | F = C−P+1 |

| Keyword | Topic | Meaning |
|---|---|---|
| rigid rotor | T4 | fixed-$r_o$ model |
| rotational constant | T4 | $B = h/8\pi^2Ic$ |
| semi-empirical | T13 | approximate Schrödinger |
| Stokes line | T2 | Raman line at lower frequency |
| thermal analysis | T12 | cooling-curve study |
| triple point | T10 | 0.0098 °C, 4.58 mmHg |
| univariant | T8 | F = 1 |
| vertical transition | T7 | Franck–Condon |
| wavenumber | T1 | $1/\lambda$ in $cm^{-1}$ |
| zero-point energy | T5 | ½ṽ_osc |

# Last 10 minutes

1. $E = h\nu$, $\bar{\nu} = 1/\lambda$
2. $E_{rot} \ll E_{vib} \ll E_{elec}$
3. $\varepsilon_J = BJ(J+1)$; lines $2B(J+1)$; spacing $2B$
4. $I = \mu r_0^2$; $r_0 = \sqrt{(I/\mu)}$
5. Rotational: permanent dipole, $\Delta J = \pm 1$
6. $\bar{\nu} = (1/2\pi c)\sqrt{(k/\mu)}$; $\varepsilon_v = (v+½)\bar{\nu}$; ZPE $= ½\bar{\nu}$
7. IR: dipole change, $\Delta v = \pm 1$ (SHO), one line
8. Anharmonic: fund $\bar{\omega_e}(1-2x_e)$, overtone $2\bar{\omega_e}(1-3x_e)$
9. Franck–Condon: vertical; 3 cases; continuum = dissociation
10. $F = C-P+2$; derivation $P(C-1)+2 - C(P-1)$
11. Water: $F=3-P$; triple 0.0098 °C, 4.58 mmHg; critical 374 °C, 218 atm
12. Pb–Ag: $F=C-P+1$; eutectic 303 °C, 2.6 % Ag, $F=0$